

Agentic Artificial Intelligence

Chapter 02: Perception, Memory, Planning & Learning

How an Agent Perceives, Remembers, and Decides

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Review + Questions

What do you remember from Chapter 1?

PEAS

Performance, Environment,
Actuators, Sensors

Taxonomy

Simple Reflex → Model →
Goal → Utility → Learning

Environment

Observable? Deterministic?
Episodic? Static? ...

Cognitive Arch.

Perception → Memory →
Planning → Action

Semester Plan

Chapter 1. Agents and their Fundamentals

Chapter 2. Perception, Memory, Planning & Learning

Chapter 3. LLMs, RAG & Tool Use

Chapter 4. Multi-Agent Systems and MARL

Chapter 5. Agentic AI Engineering

Chapter 6. Safety & Guardrails

Guiding Question

How does an agent perceive the world, remember, and decide?



The Pizza Agent

Attempt 1: No capabilities

"Order me a pizza"

Agent has no menu, no memory of preferences, no access to delivery service.

→ **Responds politely but does nothing.**

Attempt 2: Full architecture

Same request, but now with:

- Menu access (Perception)
- Past orders (Memory)
- Delivery API (Action)

→ **Orders your favourite pizza.**

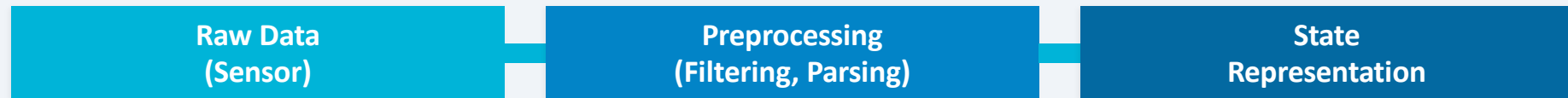
Discussion: The agent works. But imagine it has been running for 3 months. What could still go wrong — and in which layer?

PART 1

Perception

How does an agent perceive the world?

From Sensors to Representations



Key insight: Perception $\neq$ raw sensor data. It is the process of transforming raw inputs into an internal representation the agent can act upon.

System	Raw Data	Preprocessing	State Representation
Spam Filter	Email (Header + Body)	Tokenisation	Feature vector
Autonomous Car	Camera, Lidar	Object detection	3D scene graph
LLM Chatbot	Text prompt	Tokenisation	Token sequence
MARS LaserTag	Explore() return	Neighbour parsing	Local state vector

State Representation as Bottleneck

What the agent cannot represent, it cannot use.

Why does a spam filter miss phishing emails embedded as images?

Because its perception module only extracts text features. The image is there as raw data — but it is not represented in the agent's internal state.

Analogy: A person in a dark room. The world is there — but perception is insufficient to act meaningfully. The environment is "partially observable" (recall Chapter 1).

The Grounding Problem

An LLM "knows" that water boils at 100°C — but it has never experienced it.
It knows the sentence, not the phenomenon.

The Grounding Problem: The symbols a model works with are not anchored in physical reality.

Consequence for agents:

An agent built on an ungrounded LLM can produce plausible-sounding plans that are physically nonsensical. It can compute, but not understand why a glass falls off a table edge.

Discussion:

→ How could we teach an agent to understand the physical world?

Two Approaches: World Models & LQMs

World Models

The agent builds an internal simulation model to predict consequences before acting.

Instead of just "knowing" a glass can fall, it can simulate the fall.

→ Deliberative planning requires a World Model (recall: Goal-Based Agent, Ch. 1)

Large Quantitative Models

Trained on structured, numerical data using physical principles & mathematical reasoning.

An LLM summarises text about molecular bonds. An LQM simulates the bond and predicts docking — based on equations, not language patterns.

	LLM	LQM
Learns from...	Text (unstructured)	Numerical data, equations
World understanding	Linguistic correlations	Physical / mathematical principles
Grounding	Weak — symbols without anchoring	Strong — measurable quantities

PART 2

Memory

What does an agent know, and for how long?

Why Memory Matters

Without memory, every moment is the first. The agent cannot learn, plan, or respond to context.

Simple Reflex Agent

No memory

Reacts only to current percepts

Model-Based Agent

Internal model

Remembers how the world evolves

Learning Agent

Experience store

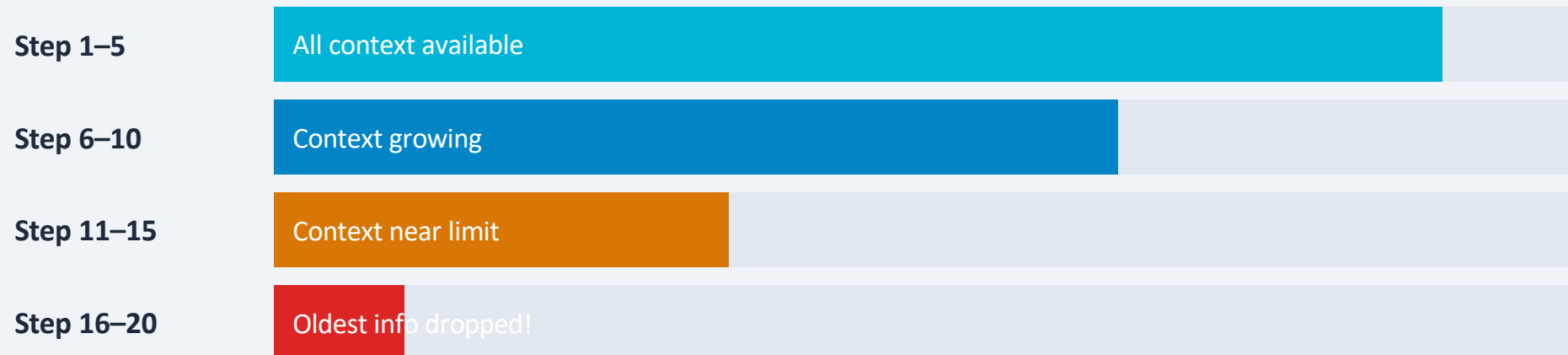
Memory as basis for improvement

Memory Taxonomy

Memory Type	Description	Human Analogy	Agent Example
Sensory / Buffer	Very short-term, current percepts	Iconic memory	LLM: current prompt
Working Memory	Active context for current task	Working memory	LLM: Context Window Agent: plan variables
Episodic Memory	Stored experiences and episodes	"I remember that..."	Chat history; action log
Semantic / LT	General knowledge, facts	World knowledge	Knowledge base; model weights; vector DB

The Context Window Problem

"An LLM has no real memory. Its working memory is the context window — and it has a hard limit."



→ How can we solve this? (Preview: RAG in Chapter 3)

Memory: ABM vs. Agentic AI

Aspect	ABM (MARS)	Agentic AI
Typical scope	Few variables per agent	Potentially unlimited context
Persistence	Across entire simulation	Often only within a session
Explicitness	Fully inspectable	Often opaque (in model weights)
Shared memory	Via environment / layers	Via external DBs / APIs

In ABM, memory is a deliberate design element — you decide exactly what an agent knows.

In Agentic AI, memory is an engineering problem — because LLMs forget by nature.

PART 3

Planning

How does an agent decide what to do?

Reactive vs. Deliberative Planning

Reactive (Sense → Act)

Direct mapping: percept → action
No internal model, no lookahead

Fast but limited

Examples: Thermostat, Simple Reflex Agent

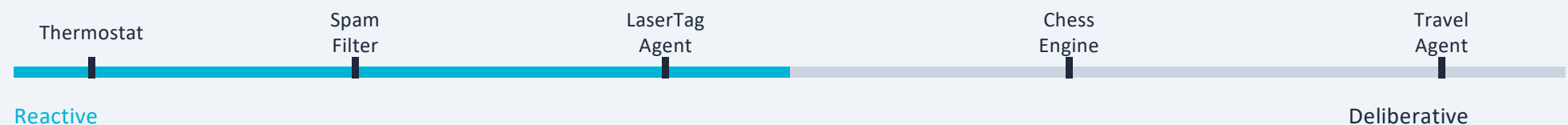
Deliberative (Sense → Think → Act)

Agent simulates consequences internally
Uses memory + model for lookahead

Slower but flexible

Examples: Chess engine, Travel agent

The Planning Spectrum



Planning Strategies

Rule-based (If-Then)

→ *Simple Reflex Agent (Ch. 1)*

Simplest form. Explicit,
transparent, easy to test.
Limited scalability.

Search-based

→ *Goal-Based Agent (Ch. 1)*

Agent searches state space for a
path to the goal.
BFS, DFS, A*, Minimax.

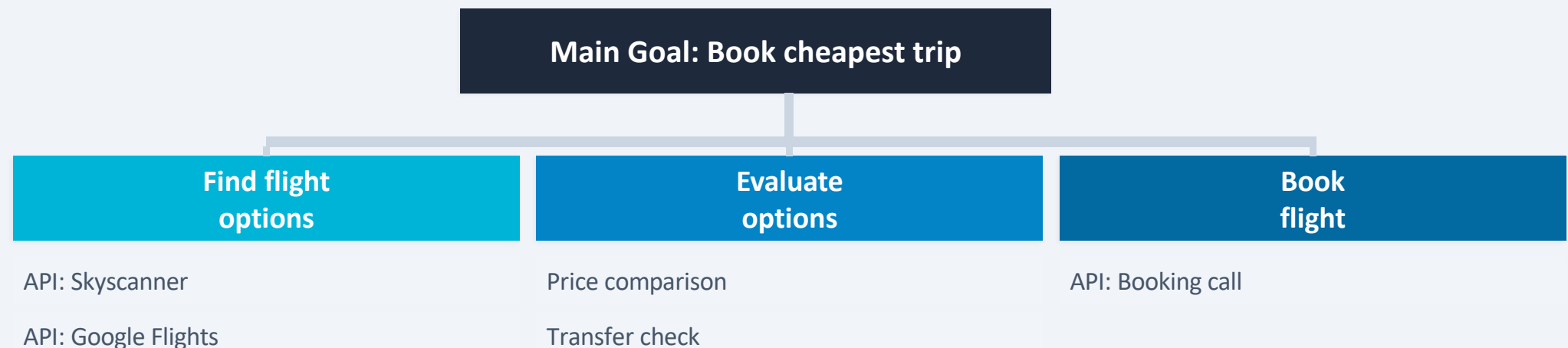
LLM-based (Prompt-as-Plan)

→ *Preview: ReAct (Ch. 3)*

Language model generates a plan
as text, then executes step by step.
Flexible but non-deterministic.

Hierarchical Planning & Decomposition

Example: "Book the cheapest trip Hamburg → Tokyo"



Key question: Who performs the decomposition — the developer or the agent itself?

PART 4

Action

How does an agent affect the world?

Three Dimensions of Action

Capability

Which actions are available?

LLM: text output only
vs. Agent with tool use

Granularity

How fine-grained?

"Steer 2° left"
vs. "Drive to Berlin"

Irreversibility

Can it be undone?

Delete an email
vs. Send an email

An LLM can formulate a perfect plan — but without actuators it remains just words.

PART 5

Putting It All Together

The four layers as a system

Example A: MARS LaserTag Agent

PERCEPTION

Explore() method → local neighbourhood (enemies, cover, teammates)

MEMORY

Few variables: Health, Ammo, last known position

PLANNING

Rule-based: If enemy_near → hide; If health_low → retreat

ACTION

Shoot(), Kneel(), Hide()

Example B: LLM-based Travel Agent



Weakness Diagnosis

What happens when a layer is missing or weak?

Weak Layer	Symptom	Example
Perception	Overlooks relevant information	Spam filter misses image-based phishing
Memory	Repeats mistakes, forgets context	LLM ignores user preferences from earlier
Planning	Chooses suboptimal actions	Thermostat heats with window open
Action	Knows what to do, but can't do it	ChatGPT without tools can't book a flight

A Quick Look Back

Perception $\neq$ Sensor data. What you don't represent doesn't exist for the agent.

Memory makes the difference between a tool and a partner.

Planning is a spectrum — from If-Then rules to LLM-generated plans.

Without Action, everything is just theory.

Preview: Chapter 3

We now know which layers an agent needs.

Next week: How do LLMs — which have neither memory nor action by default — become real agents through clever engineering?

RAG

= Memory Solution

External knowledge retrieval to overcome the context window limit

Tool Use

= Action Enabler

Function calling to bridge the gap between knowing and doing

References & Further Reading

Russell, S. & Norvig, P., "Artificial Intelligence: A Modern Approach" — Chapters Kap. 3.1–3.3 (Problem-Solving Agents, Searching) (HIBS)

Sumers, T. et al. (2024), "Cognitive Architectures for Language Agents" — arXiv:2309.02427 (see materials from chapter 1)

Park et al. (2023): "Generative Agents: Interactive Simulacra of Human Behavior" — arXiv:2304.03442